



Introduction to HTML

20m



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HTML (Hypertext Markup Language) is a markup language used for creating documents to be displayed in a web browser. Since 1990, the language has gone through several revisions and improvements, which has led to the development of several versions of HTML (W3C, 2015).

HTML5 (also known as the HTML Living Standard) is the current version, which is supported by the World Wide Web Consortium (W3C) and maintained by the Web Hypertext Application Technology Working Group (WHATWG).

The W3C is the international organisation that develops open standards for the web.

WHATWG is a community of people interested in improving HTML and includes individuals from browser vendors such as Apple and Google.

The additional features of the HTML5 specification are (W3C Working Group, 2014):^{fnref:356f2c}

- It provides a single language.
- It defines a processing model that encourages interoperable implementations (that is, compatible across major browsers and older ones).
- It improves the markup language (for example, some elements in HTML4 were deprecated, and new semantic elements were added).
- It incorporates markup and application programming interfaces (APIs) for web applications.

HTML elements

Before we get into advanced HTML, let's review HTML elements.

An HTML document is made up of a series of elements, which are used to flag or markup different parts of content for the browser. HTML elements are defined with enclosing tags that indicate how the content should appear or act.

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As we know, an HTML **element** generally consists of a start tag and an end tag with the content sandwiched in the middle.

For example:

This is an HTML element: `<h1>Web Technologies - BIT703</h1>`

- `<h1>` is the **start tag**
- Web Technologies – BIT703 is the **content or text**
- `</h1>` is the **end tag**.

In Example 5.3, hover over each part of the display text to see what they are. These follow the standard format (syntax) for an element.



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HTML

CSS

Result

EDIT ON

LIVE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
content="width=device-width,
initial-scale=1.0">
  <title>Interactive HTML Element
Diagram</title>
  <link rel="stylesheet"
href="styles.css">
</head>
<body>
  <div class="diagram">
    <div class="part opening">
      &lt;p&gt;<span
class="tooltip">Opening tag</span>
    </div>
    <div class="part content">
      My paragraph...<span
class="tooltip">Content</span>
    </div>
    <div class="part closing">
      &lt;/p&gt;<span
class="tooltip">Closing tag</span>
    </div>
  </div>
</body>
```

Resources1x0.5x0.25xRerun

Example 5.3 HTML Elements



However, some elements don't include end tags. These are referred to as empty elements (also known as self-closing or void elements). An example of an empty element is the `<br>` element, from an earlier task, that is used to insert a line break.

In Example 5.4, as you can see below, the `<br>` element does not require an end tag.
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It is the HTML5 specification that defines both the elements and their syntax.

All languages have a syntax or structure. This means that the words must be arranged in a certain way (grammar and spelling) to create a meaningful sentence. In HTML, syntax is the arrangement of elements and attributes that creates well-formed documents.

Read the HTML code below, which shows how HTML elements are used. We will refer to this code again on the following page. Remember this code as Example 5.5.



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HTML

Result

EDIT ON

LIVE

```
<!DOCTYPE html>
<html>
<html lang="en">

<head>
  <title>Distance Learning | Leading
Online Study | Open Polytechnic NZ
</title>
</head>

<body>
  <h1>Web Technologies – BIT703</h1>
  <p>This online course will give
you the knowledge and skills needed
to design and develop contemporary
data-driven
web applications.</p>
</body>

</html>
```

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Example 5.5 HTML Elements



The HTML elements that are present in Example 5.5 are explained below.

Elem ent	Description
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<code><!DOCTYPE ></code>	This is the document type used by the browser to render the page, following the standard specification. In the HTML code above, <code><!DOCTYPE html></code> means the document type is HTML5. It is important to note that it is case-sensitive.
<code><head></code>	This is the element at the top of the HTML page and includes metadata about the document.
<code><title></code>	This is the title of the page.
<code><body></code>	This includes all the contents that are visible to the user, such as paragraphs, headings and tables.
<code><h1></code>	This defines the level of heading; in this case heading 1.
<code><p></code>	This defines a paragraph.

Attributes

All HTML elements can have attributes that provide additional information to the element to control how they work. Attributes are added to the start tag and are formatted as `name = 'value'`.

For example, when using the `lang` attribute to state the language of the web page, you'd format it as:

```
<html lang = "en">
```

This code would declare the language of the web page is English (en).

The `lang` attribute always goes inside the `<html>` tag



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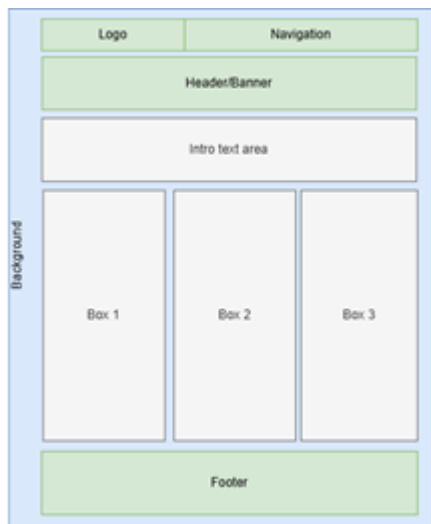
☑ Wireframe

Non-Assessed

Status: Done

Reset ↺

Consider the wireframe below and create a basic HTML structure that best reflects the page structure.



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☐ I have completed this activity



Each section should be enclosed within an HTML `<div>` tag and written in a way similar to the code in the CodePen example: Task: Wireframing.

Previous attempts ▼



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Dig deeper

Here are some additional resources about elements and attributes that you might find helpful.

- Dofactory has a list of tags and live examples of their use on the page [HTML tags](#) as well as a list of global attributes, multi-tag attributes and single-tag attributes on the page [HTML attributes](#).
- The W3Schools site has a full alphabetical list with descriptions of elements and attributes.

The next section will look at how browsers read HTML, and then we will look at the Document Object Model (DOM).



Thoughts on this page?

